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|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|
| 5   | (a) (uniform magnetic) flux normal to long (straight) wire carrying a current of 1 A<br>(creates) force per unit length of $1 \text{ N m}^{-1}$ | M1<br>A1 | [2] |
| (b) | (i) flux density $= 4\pi \times 10^{-7} \times 1.5 \times 10^3 \times 3.5$<br>$= 6.6 \times 10^{-3} \text{ T}$                                  | C1<br>A1 | [2] |
|     | (ii) flux linkage $= 6.6 \times 10^{-3} \times 28 \times 10^{-4} \times 160$<br>$= 3.0 \times 10^{-3} \text{ Wb}$                               | C1<br>A1 | [2] |
| (c) | (i) (induced) e.m.f. proportional to rate of<br>change of (magnetic) flux (linkage)                                                             | M1<br>A1 | [2] |
|     | (ii) e.m.f. $= (2 \times 3.0 \times 10^{-3}) / 0.80$<br>$= 7.4 \times 10^{-3} \text{ V}$                                                        | C1<br>A1 | [2] |